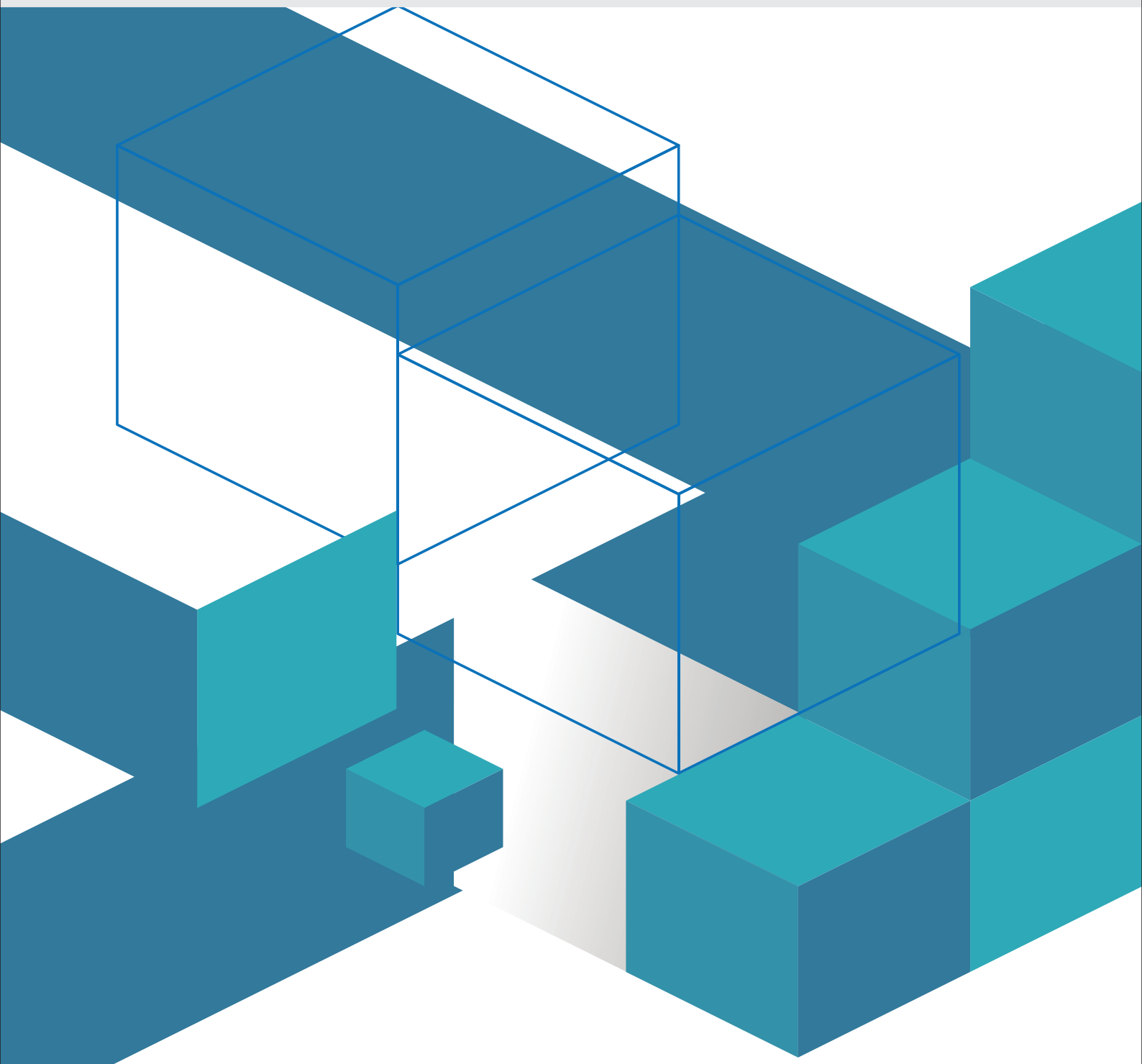
MATHEMATICS

MATRICES_MEGA_XII



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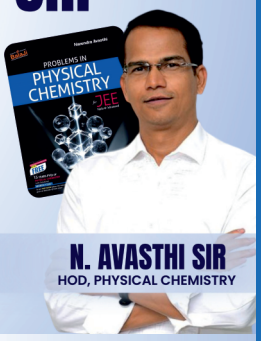

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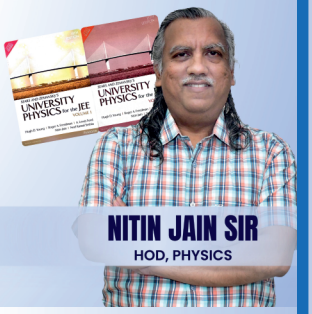


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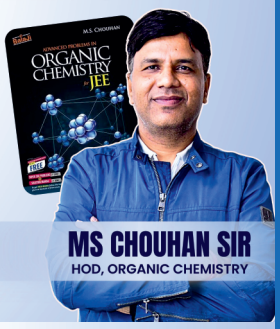


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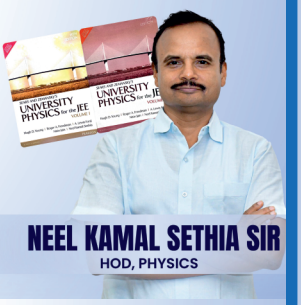


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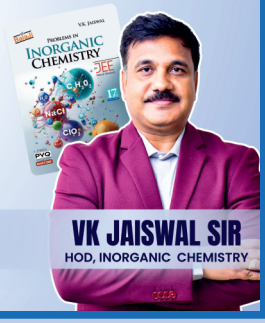


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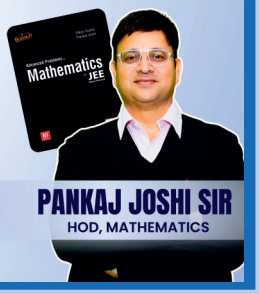


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MATHEMATICS

TARGET IIT JEE
MEGA-XII

MATRICES

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KEY CONCEPTS
MATRICES

USEFUL IN STUDY OF SCIENCE, ECONOMICS AND ENGINEERING

1. Definition : Rectangular array of m n numbers . Unlike determinants it has no value.

$$A = \begin{bmatrix} a_{11} & a_{12} & ..... & a_{1n} \\ a_{21} & a_{22} & ..... & a_{2n} \\ : & : & : & : \\ a_{m1} & a_{m2} & ..... & a_{mn} \end{bmatrix} \quad \text{or} \quad \begin{pmatrix} a_{11} & a_{12} & ..... & a_{1n} \\ a_{21} & a_{22} & ..... & a_{2n} \\ : & : & : & : \\ a_{m1} & a_{m2} & ..... & a_{mn} \end{pmatrix}$$

Abbreviated as : $A = \left[a_{ij}\right]$ $1 \le i \le m$; $1 \le j \le n$, i denotes the row and j denotes the column is called a matrix of order m × n.

2. Special Type Of Matrices :

(a) Row Matrix : $A = [a_{11}, a_{12}, ... a_{1n}]$ having one row . (1 × n) matrix.
(or row vectors)

(b) Column Matrix : $A = \begin{bmatrix} a_{11} \\ a_{21} \\ : \\ a_{m1} \end{bmatrix}$ having one column. (m × 1) matrix
(or column vectors)

(c) Zero or Null Matrix : ($A = O_{m \times n}$)

An m × n matrix all whose entries are zero .

$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ is a 3 × 2 null matrix & $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is 3 × 3 null matrix

(d) Horizontal Matrix : A matrix of order m × n is a horizontal matrix if n > m.

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 1 & 1 \end{bmatrix}$$

(e) Verical Matrix : A matrix of order m × n is a vertical matrix if m > n. $\begin{bmatrix} 2 & 5 \\ 1 & 1 \\ 3 & 6 \\ 2 & 4 \end{bmatrix}$

(f) Square Matrix : (Order n)

If number of row = number of column $\Rightarrow$ a square matrix.

Note (i) In a square matrix the pair of elements a_{ij} & a_{ji} are called **Conjugate Elements** .
e.g.

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

(ii) The elements a_{11}, a_{22}, a_{33}, ... a_{nn} are called **Diagonal Elements** . The line along which the diagonal elements lie is called **" Principal or Leading "** diagonal.

The qty Σa_{ii} = trace of the matrice written as , i.e. $t_r A$

Square Matrix

Triangular Matrix

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{pmatrix} \quad ; \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -3 & 0 \\ 4 & 3 & 3 \end{pmatrix}$$

Upper Triangular $a_{ij} = 0 \ \forall \ i > j$

Lower Triangular $a_{ij} = 0 \ \forall \ i < j$

Note that : Minimum number of zeros in a triangular matrix of order $n = n(n-1)/2$

Diagonal Matrix denote as $d_{dia}(d_1, d_2, ..., d_n)$ all elements except the leading diagonal are zero

diagonal Matrix

$$\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$$

Note: (1) If $d_1 = d_2 = d_3 = a$ Scalar Matrix

(2) If $d_1 = d_2 = d_3 = 1$ Unit Matrix

Unit or Identity Matrix

$$a_{ij} = \begin{bmatrix} 1 & \text{if} & i = j \\ 0 & \text{if} & i \neq j \end{bmatrix}$$

Note: Min. number of zeros in a diagonal matrix of order $n = n(n-1)$

"It is to be noted that with square matrix there is a corresponding determinant formed by the elements of A in the same order."

3. Equality Of Matrices :

Let $A = [a_{ij}]$ & $B = [b_{ij}]$ are equal if ,

(i) both have the same order . **(ii)** $a_{ij} = b_{ij}$ for each pair of i & j.

4. Algebra Of Matrices :

Addition : $A + B = \left[a_{ij} + b_{ij}\right]$ where A & B are of the same type. (same order)

(a) Addition of matrices is commutative.

i.e. $A + B = B + A$ $A = m \times n$; $B = m \times n$

(b) Matrix addition is associative .

$(A + B) + C = A + (B + C)$ **Note :** A , B & C are of the same type.

(c) Additive inverse.

If $A + B = \mathbf{O} = B + A$ $A = m \times n$

5. Multiplication Of A Matrix By A Scalar :

If $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$; $kA = \begin{bmatrix} ka & kb & kc \\ kb & kc & ka \\ kc & ka & kb \end{bmatrix}$

6. Multiplication Of Matrices : (Row by Column)

AB exists if , $A = m \times n$ & $B = n \times p$

2×3 3×3

AB exists , but BA does not $\Rightarrow$ $AB \neq BA$

Note : In the product AB , $\begin{cases} A = \text{prefactor} \\ B = \text{post factor} \end{cases}$

$A = (a_1, a_2, ... a_n)$ & $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

$1 \times n$ $n \times 1$

$AB = [a_1 b_1 + a_2 b_2 + ... + a_n b_n]$

If $A = \left[a_{ij}\right]$ $m \times n$ & $B = \left[b_{ij}\right]$ $n \times p$ matrix , then $\boxed{(AB)_{ij} = \sum_{r=1}^{n} a_{ir} . b_{rj}}$

Properties Of Matrix Multiplication :

1. Matrix multiplication is not commutative .

$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$; $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$; $AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$; $BA = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

$\Rightarrow$ $AB \neq BA$ (in general)

2. $AB = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $\Rightarrow$ $AB = \mathbf{O}$ $\not\Rightarrow$ $A = \mathbf{O}$ or $B = \mathbf{O}$

Note: If A and B are two non- zero matrices such that $AB = \mathbf{O}$ then A and B are called the divisors of zero.
Also if $[AB] = \mathbf{O} \Rightarrow |AB| \Rightarrow |A|\,|B| = 0 \Rightarrow |A| = 0$ or $|B| = 0$ but not the converse.
If A and B are two matrices such that

(i) $AB = BA \Rightarrow$ A and B commute each other

(ii) $AB = -BA \Rightarrow$ A and B anti commute each other

3. **Matrix Multiplication Is Associative :**
If A , B & C are conformable for the product AB & BC, then
$(A \,.\, B) \,.\, C = A \,.\, (B \,.\, C)$

4. **Distributivity :**

$$\left.\begin{aligned} A\,(B + C) &= AB + AC \\ (A + B)\,C &= AC + BC \end{aligned}\right]$$ Provided A, B & C are conformable for respective products

5. **Positive Integral Powers Of A Square Matrix :**
For a square matrix A , $A^2 A = (AA)A = A(AA) = A^3$.
Note that for a unit matrix I of any order , $I^m = I$ for all $m \in N$.

6. **MATRIX POLYNOMIAL :**
If $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots\ldots\ldots + a_nx^0$ then we define a matrix polynomial
$f(A) = a_0A^n + a_1A^{n-1} + a_2A^{n-2} + \ldots\ldots + a_nI^n$
where A is the given square matrix. If f (A) is the null matrix then A is called the zero or root of the polynomial f (x).

DEFINITIONS :

(a) **Idempotent Matrix :** A square matrix is idempotent provided $A^2 = A$.

Note that $A^n = A$ $\forall$ $n \geq 2$, $n \in N$.

(b) **Nilpotent Matrix:** A square matrix is said to be nilpotent matrix of order m, $m \in N$, if $A^m = \mathbf{O}$, $A^{m-1} \neq \mathbf{O}$.

(c) **Periodic Matrix :** A square matrix is which satisfies the relation $A^{K+1} = A$, for some positive integer K, is a periodic matrix. The period of the matrix is the least value of K for which this holds true.
Note that period of an idempotent matrix is 1.

(d) **Involutary Matrix :** If $A^2 = I$, the matrix is said to be an involutary matrix.
Note that $A = A^{-1}$ for an involutary matrix.

7. **The Transpose Of A Matrix : (Changing rows & columns)**
Let A be any matrix . Then , $A = a_{ij}$ of order $m \times n$
$\Rightarrow$ A^T or A' $= [a_{ji}]$ for $1 \leq i \leq n$ & $1 \leq j \leq m$ of order $n \times m$

Properties of Transpose : If A^T & B^T denote the transpose of A and B ,

(a) $(A \pm B)^T = A^T \pm B^T$; note that A & B have the same order.

IMP. **(b)** $(AB)^T = B^T A^T$ A & B are conformable for matrix product AB.

(c) $(A^T)^T = A$

(d) $(kA)^T = kA^T$ k is a scalar .

General : $(A_1 , A_2 , \ldots\ldots A_n)^T = A_n^T , \ldots\ldots , A_2^T , A_1^T$ (reversal law for transpose)

8. Symmetric & Skew Symmetric Matrix :

A square matrix $A = \left[a_{ij}\right]$ is said to be ,

symmetric if ,

$a_{ij} = a_{ji} \quad \forall$ i & j (conjugate elements are equal) **(Note $A = A^T$)**

Note: Max. number of distinct entries in a symmetric matrix of order n is $\frac{n(n+1)}{2}$.

and skew symmetric if ,

$a_{ij} = -a_{ji} \quad \forall$ i & j (the pair of conjugate elements are additive inverse of each other) **(Note $A = -A^T$)**

Hence If A is skew symmetric, then

$a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0 \quad \forall$ i

Thus the digaonal elements of a skew symmetric matrix are all zero , but not the converse .

Properties Of Symmetric & Skew Matrix :

P – 1 A is symmetric if $A^T = A$

A is skew symmetric if $A^T = -A$

P – 2 $A + A^T$ is a symmetric matrix

$A - A^T$ is a skew symmetric matrix .

Consider $(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$

$A + A^T$ is symmetric .

Similarly we can prove that $A - A^T$ is skew symmetric .

P – 3 The sum of two symmetric matrix is a symmetric matrix and the sum of two skew symmetric matrix is a skew symmetric matrix .

Let $A^T = A$; $B^T = B$ where A & B have the same order .

$(A + B)^T = A + B$

Similarly we can prove the other

P – 4 If A & B are symmetric matrices then ,

(a) AB + BA is a symmetric matrix

(b) AB – BA is a skew symmetric matrix .

P – 5 Every square matrix can be uniquely expressed as a sum of a symmetric and a skew symmetric matrix.

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{P} + \underbrace{\frac{1}{2}(A - A^T)}_{Q}$$

P: Symmetric Q: Skew Symmetric

9. Adjoint Of A Square Matrix :

Let $A = \left[a_{ij}\right] = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ be a square matrix and let the matrix formed by the cofactors of $[a_{ij}]$ in determinant $|A|$ is $= \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}$.

Then (adj A) $= \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$

V. Imp. Theorem : A (adj. A) = (adj. A).A = $|A|\, I_n$, If A be a square matrix of order n.

Note : If A and B are non singular square matrices of same order, then

(i) $|\text{adj } A| = |A|^{n-1}$

(ii) adj (AB) = (adj B) (adj A)

(iii) adj(KA) = K^{n-1} (adj A), K is a scalar

Inverse Of A Matrix (Reciprocal Matrix) :

A square matrix A said to be invertible (non singular) if there exists a matrix B such that,

$$AB = I = BA$$

B is called the inverse (reciprocal) of A and is denoted by A^{-1} . Thus

$$A^{-1} = B \Leftrightarrow AB = I = BA .$$

We have , $A . (\text{adj } A) = |A|\, I_n$

$$A^{-1} A (\text{adj } A) = A^{-1} I_n |A|$$

$$I_n (\text{adj } A) = A^{-1} |A| I_n$$

$$\therefore \quad A^{-1} = \frac{(\text{adj } A)}{|A|}$$

Note : The necessary and sufficient condition for a square matrix A to be invertible is that $|A| \neq 0$.

Imp. Theorem : If A & B are invertible matrices ofthe same order , then $(AB)^{-1} = B^{-1} A^{-1}$. This is reversal law for inverse.

Note :

(i) If A be an invertible matrix , then A^T is also invertible & $(A^T)^{-1} = (A^{-1})^T$.

(ii) If A is invertible, **(a)** $(A^{-1})^{-1} = A$; **(b)** $(A^k)^{-1} = (A^{-1})^k = A^{-k}$, $k \in N$

(iii) If A is an Orthogonal Matrix. $AA^T = I = A^TA$

(iv) A square matrix is said to be **orthogonal** if , $A^{-1} = A^T$.

(v) $|A^{-1}| = \frac{1}{|A|}$

SYSTEM OF EQUATION & CRITERIAN FOR CONSISTENCY

GAUSS - JORDAN METHOD

$$x + y + z = 6$$
$$x - y + z = 2$$
$$2x + y - z = 1$$

or $$\begin{pmatrix} x+y+z \\ x-y+z \\ 2x+y-z \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}$$

$$AX = B \quad \Rightarrow \quad A^{-1} A X = A^{-1} B$$

$$X = A^{-1} B = \frac{(\text{adj. } A).B}{|A|} .$$

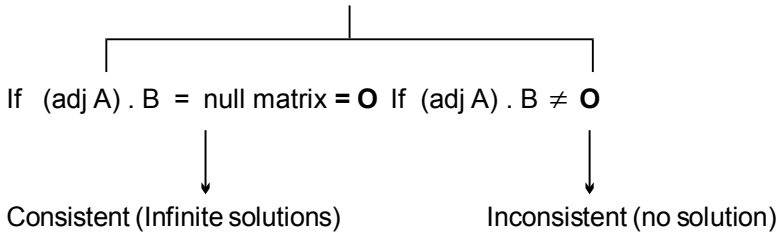
Note :

(1) If $|A| \neq 0$, system is consistent having unique solution

(2) If $|A| \neq 0$ & (adj A) . B $\neq$ **O** (Null matrix) ,
system is consistent having unique non – trivial solution .

(3) If $|A| \neq 0$ & (adj A) . B = **O** (Null matrix) ,
system is consistent having trivial solution .

(4) If $|\mathbf{A}| = 0$, **matrix method fails**



PROFICIENCY TEST-01

1. In the following, upper triangular matrix is

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & 3 \end{bmatrix}$ (B) $\begin{bmatrix} 5 & 4 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$ (D) $\begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 0 \end{bmatrix}$

2. If $A = \begin{bmatrix} 5 & 2 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then $|2A - 3B|$ equals

(A) 77 (B) –53 (C) 53 (D) –77

3. For a square matrix $A = [a_{ij}]$, $a_{ij} = 0$, when $i \neq j$, then A is

(A) unit matrix (B) scalar matrix (C) diagonal matrix (D) None of these

4. If A and B are matrices of order m × n and n × n respectively, then which of the following are defined

(A) AB, BA (B) AB, A^2 (C) A^2, B^2 (D) AB, B^2

5. If $A = \begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 5 \\ 2 & 7 \\ 3 & 10 \end{bmatrix}$, then

(A) AB and BA both exist (B) AB exists but not BA
(C) BA exists but not AB (D) Both AB and BA do not exist

6. If A is a matrix of order 3 × 4, then both AB^T and B^TA are defined if order of B is

(A) 3 × 3 (B) 4 × 4 (C) 4 × 3 (D) 3 × 4

7. Matrix $\begin{bmatrix} 0 & 5 & -7 \\ -5 & 0 & 11 \\ 7 & -11 & 0 \end{bmatrix}$ is a

(A) Diagonal matrix (B) Upper triangular matrix
(C) Skew-symmetric matrix (D) Symmetric matrix

8. If A is symmetric as well as skew symmetric matrix, then

(A) A is a diagonal matrix (B) A is a null matrix
(C) A is a unit matrix (D) A is a triangular matrix

9. If A is symmetric matrix and B is a skew-symmetric matrix, then for $n \in N$, false statement is

(A) A^n is symmetric when n is odd (B) A^n is symmetric only when n is even
(C) B^n is skew symmetric when n is odd (D) B^n is symmetric when n is even

10. Let A be a square matrix. Then which of the following is not a symmetric matrix

(A) $A + A^T$ (B) AA^T (C) A^TA (D) $A - A^T$

11. If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $n \in N$, then A^n is equal to

(A) $2^n A$ (B) $2^{n-1} A$ (B) nA (D) None of these

12. If $A = [a_{ij}]$ is scalar matrix of order $n \times n$ such that $a_{ii} = k$ for all i, then $|A|$ equals

(A) nk (B) $n + k$ (C) n^k (D) k^n

13. If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, then for every positive integer n, A^n is equal to

(A) $\begin{bmatrix} 1+2n & 4n-8 \\ n & 1-2n \end{bmatrix}$ (B) $\begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$ (C) $\begin{bmatrix} 1-2n & -4n \\ n & n-2 \end{bmatrix}$ (D) None of these

14. If A is any skew-symmetric matrix of odd orders, then $|A|$ equals

(A) –1 (B) 0 (C) 1 (D) None of these

15. If $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ and $B = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$ then AB is equal to

(A) A (B) B (C) an Identity matrix (D) a Null matrix

16. If A and B are two square matrices of the same order and $(A + B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1} B + {}^nC_2 A^{n-2} B^2 + \ldots\ldots + {}^nC_n B^n$ (n is a positive integer), then :

(A) $AB = -BA$ (B) $AB = BA$ (C) $|A| = 0$ or $|B| = 0$ (D) holds always

17. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$, $C = ABA^T$, then $A^T C^k A$ is equal to ($k \in N$, $k > 5$)

(A) $\begin{bmatrix} -k & 1 \\ 1 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} -k & 0 \\ 1 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 1 \\ -k & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ -k & 1 \end{bmatrix}$

18. If A is a skew symmetric matrix, then $B(I - A)(I + A)^{-1}$ is :

(A) Orthogonal matrix (B) Symmetric matrix
(C) Involutary matrix (D) Idempotent matrix

19. Let A be a non-singular square matrix satisfying $A^3 - 8A = 0$ then $(A^2 + I)^{-1}$ equals to :

(A) $\frac{1}{9}(9I - A^2)$ (B) $\frac{1}{9}(A^2 - 9I)$ (C) $\frac{1}{9}(9I + A^2)$ (D) $\frac{1}{9}(A^2 - I)$

20. Let A be an idempotent matrix and $(I + A)^n = I + \lambda A$ ($n \in N$) then λ equals to

(A) $2^n - 1$ (B) 2^n (C) 2^{n-1} (D) $2^n + 1$

21. Let M and N be two 3×3 non-singular symmetric matrices such that $MN = NM$. If P^T denotes the transpose of a matrix P, then $M^2N^2 (M^TN)^{-1} (MN^{-1})^T$ is equal to :

(A) $-M^2$ (B) M^2 (C) M^2N^2 (D) MN

PROFICIENCY TEST-02

1. The root of the equation $[x\ 1\ 2]\begin{bmatrix}0 & 1 & 1\\ 1 & 0 & 1\\ 1 & 1 & 0\end{bmatrix}\begin{bmatrix}x\\ -1\\ 1\end{bmatrix}=0$ is

(A) $\frac{1}{3}$ (B) $-\frac{1}{3}$ (C) 0 (D) 1

2. For square matrices A and B, AB = O, then {O : null matrix}

(A) A = O or B = O (B) A = O and B = O

(C) It is not necessary that A = O and/or B = O (D) None of these

3. If A and B are matrices of order m × n and n × m respectively, then the order of matrix $B^T (A^T)^T$ is

(A) m × n (B) m × m (C) n × n (D) Not defined

4. If $A=\begin{bmatrix}1 & 2 & 3\\ 2 & 3 & 4\\ 0 & 0 & 2\end{bmatrix}$, then the value of adj (adj A) is

(A) $4A^2$ (B) $-2A$ (C) $2A$ (D) A^2

5. If $A=\begin{bmatrix}\cos x & \sin x\\ -\sin x & \cos x\end{bmatrix}$ and A.(adj A) = $k\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then k equals

(A) sinx cosx (B) 1 (C) sin 2x (D) –1

6. If $A=\begin{bmatrix}1 & -2 & 3\\ 4 & 0 & -1\\ -3 & 1 & 5\end{bmatrix}$, then $(\text{adj } A)_{23}$ =

{i.e., the element of (adj A) which belongs to second row and third column}

(A) 13 (B) –13 (C) 5 (D) –5

7. $(\text{adj } A^T) - (\text{adj } A)^T$ equals

(A) $|A|\,I$ (B) $2|A|\,I$ (C) Null matrix (D) Unit matrix

8. If $A=\begin{bmatrix}2 & 3\\ 1 & 3\end{bmatrix}, B=\begin{bmatrix}4 & 6\\ 2 & 3\end{bmatrix}, C=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then which of these matrices are invertible ?

(A) A and B (B) B and C (C) A and C (D) All

9. Which of the following matrices is inverse of itself

(A) $\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$ (B) $\begin{bmatrix}3 & -2\\ 4 & -3\end{bmatrix}$ (C) $\begin{bmatrix}1 & 0 & 1\\ 0 & 0 & 0\\ 1 & 0 & 1\end{bmatrix}$ (D) $\begin{bmatrix}0 & 1 & 0\\ 1 & 1 & 1\\ 0 & 1 & 0\end{bmatrix}$

10. If D is a diagonal matrix with diagonal elements as $\{d_1, d_2, d_3 ..., d_n\}$ in order, then we may represent it as $D = \text{diag}(d_1, d_2, ..., d_n)$. Then D^n equals

(A) D (B) diag $(d_1^{n-1}, d_2^{n-1}, ..., d_n^{n-1})$

(C) diag $(d_1^n, d_2^n, ..., d_n^n)$ (D) None of these

11. If $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then

(A) adj A = A (B) adj A = A^{-1} (C) $A^{-1} = -A$ (D) None of these

12. If A is invertible matrix, then det (A^{-1}) equals {where, det (B) means determinant of matrix B}

(A) det (A) (B) $\frac{1}{\det(A)}$ (C) 1 (D) None of these

13. If A and B are non-zero square matrices of the same order such that AB = O, then {O : null matrix}

(A) Either adj A = O or adj B = O (B) adj A = O and adj B = O

(C) Either |A| = 0 or |B| = 0 (D) |A| = 0 and |B| = 0

14. Let A be an idempotent square matrix, then $(I + A)^4$ is :

(A) I – A (B) I + A (C) I + 15A (D) I

15. If A and B are two square matrices such that $B = -A^{-1}BA$, then $(A + B)^2$ =

(A) $A^2 + 2BA + B^2$ (B) $A^2 + B^2$ (C) $A^2 + 2AB + B^2$ (D) $A^2 - B^2$

16. If A is a diagonal matrix of order 3 × 3 is commutative with every square matrix of order 3 × 3 and trace (A) = 12 then |A| equals to

(A) 12 (B) 16 (C) 32 (D) 64

17. If A and B are two matrices of 3 × 3 such that AB = 0 and $A^2 + B = I$ then $|A^2 + B^2|$ equals to :

(A) 1 (B) –1 (C) 0 (D) 3

18. If $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, then $A^2 + A^{-1}$ =

(A) I – 3A (B) 3A – I (C) 3A + I (D) I + A

19. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 5 & 6 & 8 \end{bmatrix}$, then |adj (adj2A)| =

(A) 2^{12} (B) 2^{16} (C) 2^8 (D) 2^{32}

20. If $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, x y z = 50 and 8x + 4y + 3z = 30, then A(adj A) is equal to :

(A) 64 I (B) I (C) 48 I (D) 16 I

EXERCISE-I

1. If $5A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ x & 2 & y \end{bmatrix}$ and $AA^T = I$ then find the value of $y - x$:

2. Find the value of x and y that satisfy the equations.

$$\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} y & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$$

3. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Such that AB = B and a + d = 5050. Find the value of (ad – bc).

4. Define $A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$. Find a vertical vector V such that $(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$ (where I is the 2 × 2 identity matrix).

5. If, $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that the maxtrix A is a root of the polynomial $f(x) = x^3 - 6x^2 + 7x + 2$.

6. If the matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (a, b, c, d not all simultaneously zero) commute, find the value of $\frac{d-b}{a+c-b}$. Also show that the matrix which commutes with A is of the form $\begin{bmatrix} \alpha - \beta & 2\beta/3 \\ \beta & \alpha \end{bmatrix}$

7. If $\begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ is an idempotent matrix. Find the value of f(a), where $f(x) = x - x^2$, when bc = 1/4. Hence otherwise evaluate a.

8. If the matrix A is involutary, show that $\frac{1}{2}(I + A)$ and $\frac{1}{2}(I - A)$ are idempotent and $\frac{1}{2}(I + A) \cdot \frac{1}{2}(I - A) = \mathbf{O}$.

9. Show that the matrix $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ can be decomposed as a sum of a unit and a nilpotent marix. Hence evaluate the matrix $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}^{2007}$.

10. Find number of 2 × 2 matrices with real number as its elements satisfying $A + A^T = I$ and $AA^T = I$.

11. Let X be the solution set of the equation $A^x = I$, where $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ and I is the corresponding unit matrix and $x \subseteq N$ then find the minimum value of $\sum (\cos^x \theta + \sin^x \theta)$, $\theta \in R$.

12. $A=\begin{pmatrix}3 & a & -1\\ 2 & 5 & c\\ b & 8 & 2\end{pmatrix}$ is Symmetric and $B=\begin{pmatrix}d & 3 & a\\ b-a & e & -2b-c\\ -2 & 6 & -f\end{pmatrix}$ is Skew Symmetric, then find AB.

Is AB a symmetric, Skew Symmetric or neither of them. Justify your answer.

13. A is a square matrix of order n.
l = maximum number of distinct entries if A is a triangular matrix
m = maximum number of distinct entries if A is a diagonal matrix
p = minimum number of zeroes if A is a triangular matrix
If $l + 5 = p + 2m$, find the order of the matrix.

14. If A is an idempotent non zero matrix and I is an identity matrix of the same order, find the value of n, $n \in N$, such that $(A + I)^n = I + 127A$.

15. Consider the two matrices A and B where $A=\begin{bmatrix}1 & 2\\ 4 & 3\end{bmatrix}$; $B=\begin{bmatrix}5\\ -3\end{bmatrix}$. If n(A) denotes the number of elements in A such that n(XY) = 0, when the two matrices X and Y are not conformable for multiplication. If C = (AB)(B'A); D = (B'A)(AB) then, find the value of $\left(\frac{n(C)\left(|D|^2+n(D)\right)}{n(A)-n(B)}\right)$.

16. Let A and B are square matrices, both of order 3 × 3 satisfying $A^2 + B^4 = (A^T)^2$. Find the value of |B| (A^T is transpose matrix of A).

17. Let A and B two non-singular matrices such that $(AB)^2 = A^2B^2$ then $BA^2B^{-1} = A^n$. Find the value of n.

18. Let $A=\begin{bmatrix}\sqrt{3} & -2\\ 0 & 1\end{bmatrix}$ and P be a 2 × 2 matrix such that $PP^T = I$.

If $Q = PAP^T$ and $R = P^T Q^8 P = [r_{ij}]_{2\times 2}$. find r_{11}

19. If A and B are two square matrices of order 3 such that AB = BA and $A^2 = AB + 2B^2$. If the matrices A + B and B are non-signular, final $|AB^{-1}|$.

20. Let $A=\begin{bmatrix}-3 & 0 & 2\\ 1 & x & 5\\ -2 & 0 & x^2\end{bmatrix}$; $B=\begin{bmatrix}2\\ b\\ -1\end{bmatrix}$ and C = [3 5 1]. Find the number of integral values of 'b' for which

$tr(ABC) \le -18\ \forall\ x \in R$.

EXERCISE-II

1. $A_{3\times 3}$ is a matrix such that | A | = a, B = (adj A) such that | B | = b. Find the value of $(ab^2 + a^2b + 1)S$

where $\frac{1}{2}S = \frac{a}{b}+\frac{a^2}{b^3}+\frac{a^3}{b^5}+\ldots\ldots$ up to ∞, and a = 3.

2. For the matrix $A = \begin{bmatrix} 4 & -4 & 5 \\ -2 & 3 & -3 \\ 3 & -3 & 4 \end{bmatrix}$ find A^{-2}.

3. Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$. Find P such that $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

4. Given the matrix $A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$ and X be the solution set of the equation $A^x = A$,

where $x \in N - \{1\}$. Evaluate $\prod\left(\frac{x^3+1}{x^3-1}\right)$ where the continued product extends $\forall\ x \in X$.

5. If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then show that $F(x).\ F(y) = F(x + y)$

Hence prove that $[\ F(x)\ ]^{-1} = F(-x)$.

6. Use matrix to solve the following system of equations.

(i) $\begin{matrix} x+y+z=3 \\ x+2y+3z=4 \\ x+4y+9z=6 \end{matrix}$ (ii) $\begin{matrix} x+y+z=3 \\ x+2y+3z=4 \\ 2x+3y+4z=7 \end{matrix}$ (iii) $\begin{matrix} x+y+z=3 \\ x+2y+3z=4 \\ 2x+3y+4z=9 \end{matrix}$

7. Let A be a 3 × 3 matrix such that $a_{11} = a_{33} = 2$ and all the other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + zI$ then find the value of (x + y + z) where I is a unit matrix of order 3.

8. Find the matrix A satisfying the matrix equation, $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} . A . \begin{bmatrix} 3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & -1 \end{bmatrix}$.

9. If $A = \begin{bmatrix} k & m \\ l & n \end{bmatrix}$ and $kn \neq lm$; then show that $A^2 - (k + n)A + (kn - lm)\ I = \mathbf{O}$.

Hence find A^{-1}.

10. Given $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$. I is a unit matrix of order 2. Find all possible matrix X in the following cases.

(i) $AX = A$ (ii) $XA = I$ (iii) $XB = \mathbf{O}$ but $BX \neq \mathbf{O}$.

11. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ then, find a non-zero square matrix X of order 2 such that AX = **O**. Is XA = **O**.

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$, is it possible to find a square matrix X such that AX = O. Give reasons for it.

12. Determine the values of a and b for which the system $\begin{bmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b \\ 3 \\ -1 \end{bmatrix}$

(i) has a unique solution ; (ii) has no solution and (iii) has infinitely many solutions

13. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$; $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$ then solve the following matrix equation.

(a) AX = B – I (b) (B – I)X = IC (c) CX = A

14. If A is an orthogonal matrix and B = AP where P is a non singular matrix then show that the matrix PB^{-1} is also orthogonal.

15. Consider the matrices $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$ and let P be any orthogonal matrix and $Q = PAP^T$ and $R = P^TQ^KP$ also $S = PBP^T$ and $T = P^TS^KP$

	Column I	**Column II**
(A)	If we vary K from 1 to n then the first row first column elements at R will form	(P) G.P. with common ratio *a*
(B)	If we vary K from 1 to n then the 2nd row 2nd column elements at R will form	(Q) A.P. with common difference 2
(C)	If we vary K from 1 to n then the first row first column elements of T will form	(R) G.P. with common ratio *b*
(D)	If we vary K from 3 to n then the first row 2nd column elements of T will represent the sum of	(S) A.P. with common difference – 2.

16. Let $A = [a_{ij}]_{3\times3}$ be a matrix, where $a_{ij} = \begin{cases} i-j & ; \text{ if } i \neq j \\ i^2 & ; \text{ if } i = j \end{cases}$. If C_{ij} be the co-factors of a_{ij} in the matrix A. If B = $[b_{ij}]_{3\times3}$ be a matrix such that $b_{ij} = \sum_{k=1}^{3} a_{ik} C_{jk}$. Find the value of |B|.

17. Consider a square matrix A of order 3 such that |A| = 4. If adj(adj A) = λA, find the value of λ.

18. If A be a non-singular matrix of order 2 such that A + adjA = A^{-1}. Find the value of $|3A^{-1}|$.

19. If A and B are square matrices of the same order such that |A| = |B| = λ (λ > 0) and A(adj A + adj B) = B. Find the value of |A + B|

20. If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$. Find the value of det (A^4) + det $(A^{10} - (\text{adj}(2A))^{10})$.

EXERCISE-III

1. If $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ and $A^2 = I_2$ (2 × 2 Identity matrix), then α, β and γ will satisfy the relation

(A) $1 + \alpha^2 + \beta\gamma = 0$ (B) $1 - \alpha^2 + \beta\gamma = 0$ (C) $1 + \alpha^2 - \beta\gamma = 0$ (D) $-1 + \alpha^2 + \beta\gamma = 0$

2. If $A_\alpha = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$, then which of following statement is true

(A) $A_\alpha \cdot A_\beta = A_{\alpha\beta}$ and $(A_\alpha)^n = \begin{bmatrix} \cos^n\alpha & \sin^n\alpha \\ -\sin^n\alpha & \cos^n\alpha \end{bmatrix}$ (B) $A_\alpha \cdot A_\beta = A_{\alpha\beta}$ and $(A_\alpha)^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$

(C) $A_\alpha \cdot A_\beta = A_{\alpha+\beta}$ and $(A_\alpha)^n = \begin{bmatrix} \cos^n\alpha & \sin^n\alpha \\ -\sin^n\alpha & \cos^n\alpha \end{bmatrix}$ (D) $A_\alpha \cdot A_\beta = A_{\alpha+\beta}$ and $(A_\alpha)^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$

3. If $M = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $M^2 - \lambda M - I = 0$, then λ equals

(A) –2 (B) 2 (C) –4 (D) 4

4. If A be a matrix such that inverse of 7A is the matrix $\begin{bmatrix} -1 & 2 \\ 4 & -7 \end{bmatrix}$, then A equals

(A) $\begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 4/7 \\ 2/7 & 1/7 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 4 \\ 2 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 2/7 \\ 4/7 & 1/7 \end{bmatrix}$

5. If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $(aI + bA)^2 = A$, $(a > 0)$, then

(A) $a = b = \sqrt{2}$ (B) $a = b = \frac{1}{\sqrt{2}}$ (C) $a = b = \sqrt{3}$ (D) $a = b = \frac{1}{\sqrt{3}}$

6. If A and B are square matrices such that AB = B and BA = A, then $A^2 + B^2$ is equal to

(A) 2AB (B) 2BA (C) A + B (D) None of these

7. If $\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}\begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then

(A) $a = \sin 2\theta$, $b = -\cos 2\theta$ (B) $a = \cos 2\theta$, $b = \sin 2\theta$

(C) $a = \sin 2\theta$, $b = \cos 2\theta$ (D) $a = \cos 2\theta$, $b = -\sin 2\theta$

8. Let the matrices A and B be defined as $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 7 \\ 1 & 3 \end{bmatrix}$, then the value of determinant of matrix $(2A^7B^{-1})$, is :

(A) 2 (B) 1 (C) –1 (D) –2

9. There are two possible values of A in the solution of the matrix equation

$\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$, where A, B, C, D, E, F are real numbers. The absolute value of the difference of these two solutions, is :

(A) $\frac{13}{3}$ (B) $\frac{11}{3}$ (C) $\frac{17}{3}$ (D) $\frac{19}{3}$

10. If A is a square matrix, and B is a singular matrix of same order, then for a positive integer n, $(A^{-1}BA)^n$ equals

(A) $A^{-n}B^nA^n$ (B) $A^nB^nA^{-n}$ (C) $A^{-1}B^nA$ (D) $n(A^{-1}BA)$

EXERCISE-IV

1. If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then **[AIEEE 2003]**

(A) $\alpha = a^2 + b^2$, $\beta = ab$ (B) $\alpha = a^2 + b^2$, $\beta = 2ab$

(C) $\alpha = a^2 + b^2$, $\beta = a^2 - b^2$ (D) $\alpha = 2ab$, $\beta = a^2 + b^2$

2. Let $A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$. The only correct statement about the matrix A is : **[AIEEE 2004]**

(A) A is a zero matrix (B) $A^2 = I$

(C) A^{-1} does not exist (D) A = –I, where I is a unit matrix

3. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$. If B is the inverse of A, then α is : **[AIEEE 2004]**

(A) –2 (B) 5 (C) 2 (D) –1

4. If $A^2 - A + I = O$, then the inverse of A is : **[AIEEE 2005]**

(A) A + I (B) A (C) A – I (D) I – A

5. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction **[AIEEE 2005]**

(A) $A^n = nA - (n-1)I$ (B) $A^n = 2^{n-1}A - (n-1)I$ (C) $A^n = nA + (n-1)I$ (D) $A^n = 2^{n-1}A + (n-1)I$

6. If A and B are square matrices of size n × n such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will be always true ? **[AIEEE 2006]**

(A) A = B (B) AB = BA

(C) Either A or B is a zero matrix (D) Either A or B is an identity matrix

7. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, a, b $\in$ N. Then **[AIEEE 2006]**

(A) there cannot exists any B such that AB = BA.

(B) there exists more than one but finite numbe of B's such that AB = BA.

(C) there exists exactly one B such that AB = BA.

(D) there exists infinitely many B's such that AB = BA.

8. Let $A=\begin{bmatrix}5 & 5\alpha & \alpha\\ 0 & \alpha & 5\alpha\\ 0 & 0 & 5\end{bmatrix}$. If $|A^2| = 25$, then $|\alpha|$ equals : **[AIEEE 2007]**

(A) 5^2 (B) 1 (C) 1/5 (D) 5

9. Let A be a 2 × 2 matrix with real entries. Let I be the 2 × 2 identity matrix. Denote by tr(A), the sum of diagonal entries of A. Assume that $A^2 = I$. **[AIEEE 2008]**

Statement 1 : If $A \neq I$ and $A \neq -I$, then detA = –1.

Statement 2 : If $A \neq I$ and $A \neq -I$, then $tr(A) \neq 0$.

(A) Statement 1 is false, statement 2 is true.

(B) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.

(C) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1.

(D) Statement 1 is true, statement 2 is false.

10. Let A be a 2 × 2 matrix. **[AIEEE 2009]**

Statement 1 : adj.(adj A) = A

Statement 2 : |adj A| = |A|

(A) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.

(B) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1.

(C) Statement 1 is true, statement 2 is false.

(D) Statement 1 is false, statement 2 is true.

11. The number of 3 × 3 non-singular matrices with four entries as 1 and all other entries as 0 is : **[AIEEE 2010]**

(A) at least 7 (B) less than 4 (C) 5 (D) 6

12. Let A be a 2 × 2 matrix with non-zero entries and let $A^2 = I$, where I is a 2 × 2 identity matrix. Define Tr(A) = sum of diagonal elements of A and |A| = determinant of matrix A. **[AIEEE 2010]**

Statement 1 : Tr(A) = 0

Statement 2 : |A| = 1

(A) Statement 1 is false, statement 2 is true.

(B) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.

(C) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1.

(D) Statement 1 is true, statement 2 is false.

13. Let A and B two symmetric matrices of order 3. **[AIEEE 2011]**

Statement 1 : A(BA) and (AB)A are symmetric matrices

Statement 2 : AB is symmetric matrix if matrix multiplication of A with B is commutative.

(A) Statement 1 is false, statement 2 is true.

(B) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.

(C) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1.

(D) Statement 1 is true, statement 2 is false.

14. Let $A=\begin{pmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 3 & 2 & 1\end{pmatrix}$. If u_1 and u_2 are column matrices such that $Au_1=\begin{pmatrix}1\\0\\0\end{pmatrix}$ and $Au_2=\begin{pmatrix}0\\1\\0\end{pmatrix}$, then $u_1 + u_2$ is equal to : **[AIEEE 2012]**

(A) $\begin{pmatrix}-1\\1\\0\end{pmatrix}$ (B) $\begin{pmatrix}-1\\1\\-1\end{pmatrix}$ (C) $\begin{pmatrix}-1\\-1\\0\end{pmatrix}$ (D) $\begin{pmatrix}1\\-1\\-1\end{pmatrix}$

15. Let P and Q be 3 × 3 matrices $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$, then determinant of $(P^2 + Q^2)$ is equal to : **[AIEEE 2012]**

(A) –2 (B) 1 (C) 0 (D) –1

16. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3 × 3 matrix A and $|A| = 4$, then α is equal to :

(A) 0 (B) 4 (C) 11 (D) 5 **[JEE Main - 2013]**

17. If A is an 3 × 3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1}A'$, then BB' equals **[JEE Main - 2014]**

(A) $(B^{-1})'$ (B) I + B (C) I (D) B^{-1}

18. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3 × 3 identity matrix, then the ordered pair (a, b) is equal to: **[JEE Main - 2015]**

(A) (–2, –1) (B) (2, –1) (C) (–2, 1) (D) (2, 1)

19. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and A adj $A = AA^T$, then 5a + b is equal to : **[JEE Main- 2016]**

(A) –1 (B) 5 (C) 4 (D) 13

20. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then adj $(3A^2 + 12A)$ is equal to : **[JEE Main - 2017]**

(A) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$ (B) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$ (C) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$ (D) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$

21. Let A be a matrix such that $A \cdot \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ is a scalar matrix and $|3A| = 108$. Then A^2 equals : **[JEE Main - 2018]**

(A) $\begin{bmatrix} 4 & -32 \\ 0 & 36 \end{bmatrix}$ (B) $\begin{bmatrix} 36 & 0 \\ -32 & 4 \end{bmatrix}$ (C) $\begin{bmatrix} 4 & 0 \\ -32 & 36 \end{bmatrix}$ (D) $\begin{bmatrix} 36 & -32 \\ 0 & 4 \end{bmatrix}$

22. Suppose A is any 3 × 3 non-singular matrix and (A – 3I) (A – 5I) = O, where $I = I_3$ and $O = O_3$. If $\alpha A + \beta A^{-1} = 4I$, then $\alpha + \beta$ is equal to : **[JEE Main - 2018]**

(A) 8 (B) 7 (C) 13 (D) 12

23. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = A^{20}$. Then the sum of the elements of the first column of B is **[JEE Main - 2018]**

(A) 211 (B) 210 (C) 231 (D) 251

EXERCISE-V

Paragraph for question nos. 1 to 3

Let a, b and c be three real numbers satisfying $[a\ b\ c]\begin{bmatrix}1 & 9 & 7\\ 8 & 2 & 7\\ 7 & 3 & 7\end{bmatrix} = [0\ 0\ 0]$ .....(E) **[JEE-2011]**

1. If the point P(a, b, c), with reference to (E), lies on the plane 2x + y + z = 1, then the value of 7a + b + c is
(A) 0 (B) 12 (C) 7 (D) 6

2. Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. If a = 2 with b and c satisfying (E), then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to
(A) –2 (B) 2 (C) 3 (D) –3

3. Let b = 6, with a and c satisfying (E). If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then $\sum_{n=0}^{\infty}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^n$ is
(A) 6 (B) 7 (C) 6/7 (D) ∞

4. Let M and N be two 3 × 3 non-singular skew symmetric matrices such that MN = NM. If P^T denotes the transpose of P, then $M^2N^2 (M^TN)^{-1} (MN^{-1})^T$ is equal to **[JEE-2011]**
(A) M^2 (B) $-N^2$ (C) $-M^2$ (D) MN

5. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form $\begin{bmatrix}1 & a & b\\ \omega & 1 & c\\ \omega^2 & \omega & 1\end{bmatrix}$, where each of a, b, and c is either ω or ω^2. Then the number of distinct matrices in the set S is
(A) 2 (B) 6 (C) 4 (D) 8 **[JEE-2011]**

6. Let M be a 3 × 3 matrix satisfying **[JEE-2011]**

$$M\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}-1\\2\\3\end{bmatrix}, M\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}1\\1\\-1\end{bmatrix}, \text{ and } M\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}0\\0\\12\end{bmatrix}.$$

Then the sum of the diagonal entries of M is :

7. Let $P = [a_{ij}]$ be a 3 × 3 matrix and let $Q = [b_{ij}]$ where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \le i,j \le 3$. If the determinant of P is 2, then the determinant of the matrix Q is **[JEE-2012]**
(A) 2^{10} (B) 2^{11} (C) 2^{12} (D) 2^{13}

8. If P is a 3 × 3 matrix such that $P^T = 2P + I$, where P^T is the transposes of P and I is the 3 × 3 identity matrix, then there exists a column matrix $X = \begin{bmatrix}x\\y\\z\end{bmatrix} \neq \begin{bmatrix}0\\0\\0\end{bmatrix}$ such that **[JEE-2012]**

(A) $PX = \begin{bmatrix}0\\0\\0\end{bmatrix}$ (B) PX = X (C) PX = 2X (D) PX = – X

9. If the adjoint of a 3 × 3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is (are)

(A) – 2 (B) – 1 (C) 1 (D) 2 **[JEE-2012]**

10. For 3 × 3 matrices M and N, which of the following statement(s) is (are) NOT correct?

(A) N^T M N is symmetric or skew symmetric, according as M is symmetric or skew symmetric

(B) M N – N M is skew symmetric for all symmetric matrices M and N

(C) M N is symmeric for all symmetric matrices M and N

(D) (adj M) (adj N) = adj (M N) for all invertible matrices M and N **[JEE Advanced - 2013]**

11. Let M be a 2 × 2 symmetric matrix with integer entries. Then M is invertible if

(A) the first column of M is the transpose of the second row of M **[JEE Advanced 2014]**

(B) the second row of M is the transpose of the first column of M

(C) M is a diagonal matrix with non-zero entries in the main diagonal

(D) the product of entries in the main diagonal of M is not thesquare of an integer

12. Let M and N be two 3 × 3 matrices such that MN = NM. Further, if $M \neq N^2$ and $M^2 = N^4$, then

(A) determinant of $(M^2 + MN^2)$ is 0 **[JEE Advanced 2014]**

(B) there is a 3 × 3 non-zero matrix U such that $(M^2 + MN^2)U$ is the zero matrix

(C) determinant of $(M^2 + MN^2) \geq 1$

(D) for a 3 × 3 matrix U, if $(M^2 + MN^2)U$ euqals the zero matrix then U is the zero matrix

13. Let X and Y be two arbitrary, 3 × 3, non-zero, skew-symmetric matrices and Z be an arbitrary 3 × 3, non-zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric? **[JEE Advanced 2015]**

(A) $Y^3Z^4 - Z^4Y^3$ (B) $X^{44} + Y^{44}$ (C) $X^4Z^3 - Z^3X^4$ (D) $X^{23} + Y^{23}$

14. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in$ **R.** Suppose $Q = [q_{ij}]$ is a matrix such that PQ = kI, where $k \in$ **R**, $k \neq 0$ and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then **[JEE Advanced 2016]**

(A) $\alpha = 0, k = 8$ (B) $4\alpha - k + 8 = 0$

(C) $\det(P\,\text{adj}(Q)) = 2^9$ (D) $\det(Q\,\text{adj}(P)) = 2^{13}$

15. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}}$ equals **[JEE Advanced 2016]**

(A) 52 (B) 103 (C) 201 (D) 205

16. For a real number α, if the system $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ of linear equations, has infinitely many solutions, then $1 + \alpha + \alpha^2 =$ **[JEE Advanced 2017]**

17. Which of the following is (are) NOT the square of a 3 × 3 matrix with real entries ? **[JEE-Advanced-2017]**

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

18. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_2 \in \mathbb{R}$ and the system of equations (in real variables)

$$-x + 2y + 5z = b_1$$
$$2x - 4y + 3z = b_2$$
$$x - 2y + 2z = b_3$$

has at least one solution. Then, which of the following system(s) (in real variables) has (have) at least one solution for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$? **[JEE Advanced 2018]**

(A) $x + 2y + 3z = b_1$, $4y + 5z = b_2$ and $x + 2y + 6z = b_3$

(B) $x + y + 3z = b_1$, $5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$

(C) $-x + 2y - 5z = b_1$, $2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$

(D) $x + 2y + 5z = b_1$, $2x + 3z = b_2$ and $x + 4y - 5z = b_3$

19. Let P be a matrix of order 3 × 3 such that all the entries in P are from the set {–1, 0, 1}. Then, the maximum possible value of the determinant of P is _____. **[JEE Advanced 2018]**

20. Let $M = \begin{bmatrix} \sin^4\theta & -1-\sin^2\theta \\ 1+\cos^2\theta & \cos^4\theta \end{bmatrix} = \alpha I + \beta M^{-1}$

where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real numbers, and I is the 2 × 2 identity matrix. If

α^* is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and

β^* is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi)\}$,

then the value of $\alpha^* + \beta^*$ is : **[JEE Advanced 2019]**

(A) $-\frac{29}{16}$ (B) $-\frac{37}{16}$ (C) $-\frac{17}{16}$ (D) $-\frac{31}{16}$

21. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and adj $M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$

where a and b are real numbers. Which of the following options is/are correct? **[JEE Advanced 2019]**

(A) $(\text{adj } M)^{-1} + \text{adj } M^{-1} = -M$ (B) If $M\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\alpha - \beta + \gamma = 3$

(C) $\det(\text{adj } M^2) = 81$ (D) $a + b = 3$

22. Let $x \in R$ and let **[JEE Advanced 2019]**

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix} \text{ and } R = PQP^{-1}.$$

Then which of the following options is/are correct?

(A) There exists a real number x such that PQ = QP

(B) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$, for all $x \in R$

(C) For x = 0, if $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$, then a + b = 5

(D) For x = 1, there exists a unit vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ for which $R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

23. Let

$$P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

and $X = \sum_{k=1}^{6} P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$ **[JEE Advanced 2019]**

where P_k^T denotes the transpose of the matrix P_k. Then which of the following options is/are correct?

(A) X – 30I is an invertible matrix

(C) X is a symmetric matrix

(C) If $X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\alpha = 30$

(D) The sum of diagonal entries of X is 18

24. Let M be a 3 × 3 invertible matrix with real entries and let I denote the 3 × 3 identity matrix. If $M^{-1} = \text{adj}(\text{adj } M)$, then which of the following statements is/are ALWAYS TRUE ? **[JEE Advanced 2020]**

(A) M = I (B) det M = 1 (C) $M^2 = I$ (D) $(\text{adj } M)^2 = I$

25. The trace of a square matrix is defined to be the sum of its diagonal entries. If A is a 2 × 2 matrix such that the trace of A is 3 and the trace of A^3 is – 18, then the value of the determinant of A is _____

[JEE Advanced 2020]

26. For any 3 × 3 matrix M, let |M| denote the determinant of M. Let

$$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$$

If Q is a nonsingular matrix of order 3 × 3, then which of the following statements is (are) **TRUE** ?

[JEE Advanced 2021]

(A) F = PEP and $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $|EQ + PFQ^{-1}] = |EQ| + |PFQ^{-1}|$

(C) $|(EF)^3| > |EF|^2$

(D) Sum of the diagonal entries of $P^{-1}EP + F$ is equal to the sum of diagonal entries of $E + P^{-1}FP$

27. For any 3 × 3 matrix M, let |M| denote the determinant of M. Let I be the 3 × 3 identity matrix. Let E and F be two 3 × 3 matrices such that (I – EF) is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) **TRUE**? **[JEE Advanced 2021]**

(A) |FE| = |I – FE||FGE|

(B) (I – FE)(I + FGE) = I

(C) EFG = GEF

(D) (I – FE)(I – FG) = I

28. If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to M^{2022}? **[JEE Advanced 2022]**

(A) $\begin{pmatrix} 3034 & 3033 \\ -3033 & -3032 \end{pmatrix}$ (B) $\begin{pmatrix} 3034 & -3033 \\ 3033 & -3032 \end{pmatrix}$ (C) $\begin{pmatrix} 3033 & 3032 \\ -3032 & -3031 \end{pmatrix}$ (D) $\begin{pmatrix} 3032 & 3031 \\ -3031 & -3030 \end{pmatrix}$

29. Let β be a real number. Consider the matrix

$$A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$$

If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of 9β is _____. **[JEE Advanced 2022]**

30. Let $M = (a_{ij})$, $i, j \in \{1, 2, 3\}$, be then 3 × 3 matrix such that $a_{ij} = 1$ if j + 1 is divisible by i, otherwise $a_{ij} = 0$. Then which of the following statements is(are) true? **[JEE Advanced 2023]**

(A) M is invertible

(B) There exists a nonzero column matrix $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ such that $M \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} -a_1 \\ -a_2 \\ -a_3 \end{pmatrix}$

(C) The set $\{X \in \mathbf{R}^3 : MX = 0\} \neq \{0\}$, where $0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

(D) The matrix (M – 2I) is invertible, where I is the 3 × 3 identify matrix

31. Let $R = \left\{\begin{pmatrix} a & 3 & b \\ c & 2 & d \\ 0 & 5 & 0 \end{pmatrix} : a,b,c,d \in \{0,3,5,7,11,13,17,19\}\right\}$. Then the number of invertible matices in R is :

[JEE Advanced 2023]

32. Let α and β be the distinct roots of the equation $x^2 + x - 1 = 0$. Consider the set $T = \{1, \alpha, \beta\}$. For a 3×3 matrix $M = \{a_{ij}\}_{3\times3}$, define $R_i = a_{i1} + a_{i2} + a_{i3}$ and $C_j = a_{1j} + a_{2j} + a_{3j}$ for i = 1, 2, 3 and j = 1, 2, 3.
Match each entry if **List -I** to the correct entry in **List-II**. **[JEE Advanced 2024]**

	List-I		**List-II**
(P)	The number of matrices $M = (a_{ij})_{3\times3}$ with all entries in T such that $R_i = C_j = 0$ for all i, j, is	(1)	1
(Q)	The number of symmetric matrices $M = (a_{ij})_{3\times3}$ with all entries in T such that $C_j = 0$ for all j, is	(2)	12
(R)	Let $M = (a_{ij})_{3\times3}$ be a skew symmetric matrix such that $a_{ij} \in T$ for $i > j$. Then the number of elements in the set $\left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix} : x,y,z \in \mathbf{R}, M\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_{12} \\ 0 \\ -a_{23} \end{pmatrix}\right\}$ is	(3)	infinite
(S)	Let $M = (a_{ij})_{3\times3}$ be a matrix with all entries in T such that $R_i = 0$ for all i. Then the absolute value of the determinant of M is	(4)	6
		(5)	0

The correct options is :

(A) (P) $\to$ (4), (Q) $\to$ (2), (R) $\to$ (5), (S) $\to$ (1)
(B) (P) $\to$ (2), (Q) $\to$ (4), (R) $\to$ (1), (S) $\to$ (5)
(C) (P) $\to$ (2), (Q) $\to$ (4), (R) $\to$ (3), (S) $\to$ (5)
(D) (P) $\to$ (1), (Q) $\to$ (5), (R) $\to$ (3), (S) $\to$ (4)

ANSWER KEY

PROFICIENCY TEST-01

1.	B	**2.**	B	**3.**	C	**4.**	D	**5.**	A	**6.**	D	**7.**	C
8.	B	**9.**	B	**10.**	D	**11.**	B	**12.**	D	**13.**	B	**14.**	B
15.	D	**16.**	B	**17.**	D	**18.**	A	**19.**	A	**20.**	A	**21.**	B

PROFICIENCY TEST-02

1.	A	**2.**	C	**3.**	D	**4.**	B	**5.**	B	**6.**	A	**7.**	C
8.	C	**9.**	B	**10.**	C	**11.**	B	**12.**	B	**13.**	D	**14.**	C
15.	B	**16.**	D	**17.**	A	**18.**	B	**19.**	A	**20.**	C		

EXERCISE-I

1. 1.00 **2.** $x = \frac{3}{2}$, $y = 2$ **3.** 5049 **4.** $V = \begin{bmatrix} 0 \\ \\ \frac{1}{11} \end{bmatrix}$ **6.** 1

7. f (a) = 1/4, a = 1/2 **9.** $\begin{bmatrix} 1 & 0 \\ 4014 & 1 \end{bmatrix}$ **10.** 2.00

11. 2 **12.** AB is neither symmetric nor skew symmetric

13. 4 **14.** n = 7 **15.** 650 **16.** 0.00

17. 2.00 **18.** 81.00 **19.** 8.00 **20.** 5.00

EXERCISE-II

1. 225 **2.** $\begin{bmatrix} 17 & 4 & -19 \\ -10 & 0 & 13 \\ -21 & -3 & 25 \end{bmatrix}$ **3.** $\begin{bmatrix} -4 & 7 & -7 \\ 3 & -5 & 5 \end{bmatrix}$ **4.** 3/2

6. (i) $x = 2, y = 1, z = 0$; (ii) $x = 2 + k,\ y = 1 - 2k,\ z = k$ where $k \in R$;

(iii) inconsistent, hence no solution

7. 1 **8.** $\frac{1}{19}\begin{bmatrix} 48 & -25 \\ -70 & 42 \end{bmatrix}$ **9.** $\frac{1}{kn - \ell m}\begin{bmatrix} n & -m \\ -\ell & k \end{bmatrix}$

10. (i) $X = \begin{bmatrix} a & b \\ 2-2a & 1-2b \end{bmatrix}$ for $a, b \in R$; (ii) X does not exist ;

(iii) $X = \begin{bmatrix} a & -3a \\ c & -3c \end{bmatrix}$ $a, c \in R$ and $3a + c \neq 0$; $3b + d \neq 0$

11. $X = \begin{bmatrix} -2c & -2d \\ c & d \end{bmatrix}$, where $c, d \in R - \{0\}$, NO

12. (i) $a \neq -3$, $b \in R$; (ii) $a = -3$ and $b \neq 1/3$; (iii) $a = -3$, $b = 1/3$

13. (a) $X = \begin{bmatrix} -3 & -3 \\ 5/2 & 2 \end{bmatrix}$, (b) $X = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, (c) no solution **15.** (A) Q; (B) S; (C) P; (D) P

16. $(62)^3$ **17.** 4.00 **18.** 18.00 **19.** λ^{2-n} **20.** 16.00

EXERCISE-III

1.	D	**2.**	D	**3.**	D	**4.**	D	**5.**	B	**6.**	C	**7.**	B
8.	D	**9.**	D	**10.**	C								

EXERCISE-IV

1.	B	**2.**	B	**3.**	B	**4.**	D	**5.**	A	**6.**	B	**7.**	D
8.	C	**9.**	D	**10.**	B	**11.**	A	**12.**	D	**13.**	C	**14.**	D
15.	C	**16.**	C	**17.**	C	**18.**	A	**19.**	B	**20.**	D	**21.**	D
22.	A	**23.**	C										

EXERCISE-V

1.	D	**2.**	A	**3.**	B	**4.**	**Bonus**	**5.**	A
6.	9	**7.**	D	**8.**	D	**9.**	A, D	**10.**	C, D
11.	C, D	**12.**	A,B	**13.**	C,D	**14.**	B, C	**15.**	B
16.	1	**17.**	A,C	**18.**	A, D	**19.**	4	**20.**	A
21.	ABD	**22.**	BC	**23.**	BCD	**24.**	BCD	**25.**	5.00
26.	A,B,D	**27.**	A,B,C	**28.**	A	**29.**	3.00	**30.**	B,C
31.	3780.00	**32.**	C						